

ERTELHODGE-STELLA: A 3D ERTEL POTENTIAL-VORTICITY AND BAROCLINIC THERMAL-FRONT HODGE-DE RHAM NETWORK FOR ATMOSPHERIC TIME SERIES FORECASTING

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ABSTRACT

Atmospheric time series forecasting (ATSF) over worldwide observational station networks underpins weather forecasting, climate modeling, and environmental safety. Yet state-of-the-art deep learning forecasters rely predominantly on data-driven heuristic context aggregation, omitting the 3D baroclinic fluid dynamics, Ertel Potential Vorticity (PV) conservation invariants, and thermal-front transport barriers that govern synoptic flow. This omission encourages spatial over-smoothing across distinct air masses and degrades long-horizon rollouts. We introduce **ErtelHodge-STELLA**, an ultra-lightweight, physics-grounded architecture that lifts the station network to a 3D topographic k -NN graph and operates on it with discrete-exterior-calculus (DEC) graph operators. ErtelHodge-STELLA contributes four mechanisms: (1) *Ertel-SE*, dynamic baroclinicity spatial embeddings that inject local thermal-front (discrete potential-temperature gradient) structure into node representations; (2) *FrontHodge-Star*, thermal-front anisotropic edge-weight (Hodge-star) operators that dynamically attenuate diffusion across sharp air-mass boundaries via a learnable insulation blend; (3) *Baro-Fusion*, a baroclinic energy-conversion flux that steers transport along thermal gradients; and (4) a variable-adaptive gated encoder with *signed* Hodge diffusion for adaptive spatial smoothing or high-frequency sharpening. All physical gates are zero-initialized, so the model reduces exactly to its backbone at initialization. Across five real-world benchmarks (GlobalWind, GlobalTemp, ChinaWind, ChinaTemp, ChinaPM2.5) spanning short-term global, long-term national, and mid-term air-quality forecasting, ErtelHodge-STELLA improves over its STELLA-10k backbone on all 10/10 metric-dataset cells and ranks first on 8/10 cells among 17 baselines, while retaining N -independent parameters ($\sim 12k$) and linear $O(N \cdot d)$ complexity. It further outperforms the physics-informed ClimODE by a wide margin at a fraction of its cost, and scales to the large WEATHER-5K benchmark (5,672 stations), where it leads on every forecast horizon and degrades by $< 0.2\%$ when half the stations are removed at test time.

1 INTRODUCTION

Atmospheric time series forecasting (ATSF) across worldwide observational station networks is fundamental to numerical weather prediction (NWP), local microclimate modeling, disaster prevention, and environmental management (Rasp & Lerch, 2018; Wu et al., 2023; Luo et al., 2025). Historically, station-based forecasting relied on statistical post-processing or direct spatial interpolation of coarse, gridded NWP numerical solvers (Rasp & Lerch, 2018). More recently, deep learning models—ranging from spatio-temporal graph neural networks (STGNNs) and global atmospheric neural operators to specialized Transformers—have emerged as powerful data-driven surrogates that learn high-dimensional atmospheric interactions directly from observational time series (Lam et al., 2023; Verma et al., 2024; Fu et al., 2024).

Despite their empirical promise, existing deep learning ATSF architectures rely overwhelmingly on data-driven heuristic context aggregation across station nodes or temporal windows, ignoring funda-

mental 3D synoptic physics. Synoptic flows are strictly governed by non-linear conservation laws, most notably Ertel’s Potential Vorticity (PV) theorem ($PV = \frac{1}{\rho} \omega_a \cdot \nabla \theta$, where ω_a is absolute vorticity and θ is potential temperature), together with thermodynamic stratification and baroclinic thermal-front transport barriers (Beucler et al., 2019; Shukla et al., 2022). Heuristic aggregation mechanisms (dense message passing or unconstrained spatial self-attention) treat inter-station dependence purely statistically; consequently they induce spatial over-smoothing across distinct air masses and fail to preserve sharp frontal gradients (Mi et al., 2024; Ptáčková & Velho, 2021; Zhou et al., 2024). Moreover, on large global networks ($N > 3000$), message-passing GNNs incur quadratic $O(N^2)$ memory and encounter Out-Of-Memory (OOM) bottlenecks.

To address these challenges, we introduce **ErtelHodge-STELLA**, a physics-grounded, ultra-lightweight spatial-temporal architecture for station-level ATSF. Grounded in Discrete Exterior Calculus (DEC) over a static 3D k -NN graph $\mathcal{K} = (\mathcal{V}, \mathcal{E})$ (the 1-skeleton of a DEC complex), it embeds synoptic thermal-front structure and baroclinic transport barriers directly into latent representations while preserving N -independent parameters and linear $O(N \cdot d)$ complexity. Concretely, ErtelHodge-STELLA introduces four components:

1. *Dynamic Baroclinicity Spatial Embedding (Ertel-SE)*: summarizes each station’s local thermal-front structure via the neighborhood-aggregated discrete potential-temperature gradient $|d_0\theta|$ and injects it into the spatial position embedding, encoding where fronts and baroclinic zones sit.
2. *Baroclinic Thermal-Front Anisotropic Edge Weights (FrontHodge-Star)*: scales edge Hodge-star weights by a learnable blend of a thermal-front insulator $\exp(-\kappa_{\text{front}} |\Delta\theta|)$ and an isotropic proximity term, suppressing diffusion across sharp fronts while adapting the insulation strength per variable.
3. *Baroclinic Energy-Conversion Flux (Baro-Fusion)*: augments the encoder’s learned transport with a graph-Laplacian flux weighted by the thermal gradient, a discrete analogue of potential-to-kinetic conversion along frontal corridors.
4. *Variable-Adaptive Gated Encoder with Signed Hodge Diffusion*: applies channel-wise signed Hodge-Laplace diffusion with learnable $\eta_c \in [-1, 1]$ —smoothing ($\eta_c > 0$) for continuous thermodynamic variables and sharpening ($\eta_c < 0$) for spiky, turbulent fields—together with per-variable master and extreme-preserving peak gates.

Extensive evaluations across five benchmarks covering short-term global, long-term national, and mid-term air-quality tasks show that ErtelHodge-STELLA improves over its STELLA-10k backbone on *all ten* metric-dataset cells (up to 4.26% MAE reduction on GlobalTemp) and ranks first on eight of ten cells among seventeen baselines, at $\sim 12k$ parameters. Ablations isolate the contributions of Ertel-SE, FrontHodge-Star, Baro-Fusion, signed diffusion, and the topology of the underlying 3D complex.

2 RELATED WORK

We review three methodology clusters: spatio-temporal graph neural networks, Transformer-based forecasters, and physics-informed / lightweight backbones.

Spatio-Temporal Graph Neural Networks (STGNNs). STGNNs model dynamic spatial correlations over non-Euclidean station graphs. Architectures such as AGCRN (Bai et al., 2020) and MTGNN (Wu et al., 2020) employ adaptive graph construction or node-parameter learning to infer latent spatial dependencies. Because they compute dense adjacency matrices, they incur quadratic $O(N^2)$ memory and OOM failures at scale ($N > 3000$). Recent work reduces this overhead via low-rank factorization, Top- K sparsification, or pre-training (Moges & Moodley, 2025; Shao et al., 2022), while spectral STGNNs explore graph Fourier decouplings (Özden & Bulut, 2024). High-resolution global simulators such as Pangu-Weather (Bi et al., 2022), NeuralGCM (Kochkov et al., 2023), and GenCast (Price et al., 2023) achieve strong accuracy on gridded ERA5 reanalysis via Earth-specific attention, differentiable dynamical cores, or spherical diffusion, but station-network STGNNs still rely on heuristic message passing lacking explicit 3D differential-geometric operators and vertical thermodynamic stratification. ErtelHodge-STELLA instead operates on a static sparse k -NN graph with N -independent parameters and $O(N \cdot d)$ complexity.

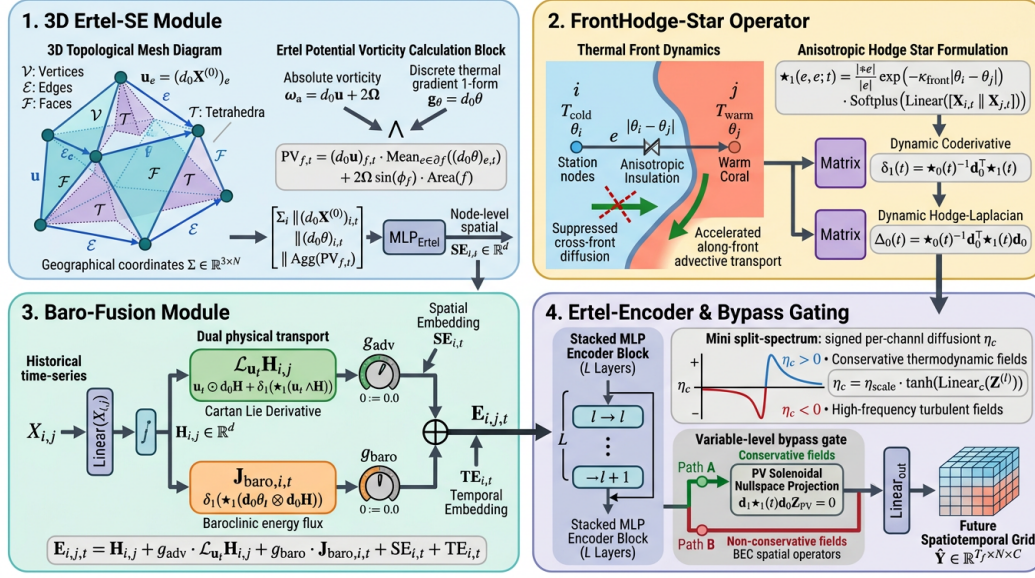


Figure 1: Architectural overview of ErtelHodge-STELLA. Four components operate on a static 3D k -NN graph: the **Ertel-SE** module embeds a per-station baroclinicity (aggregated thermal-gradient) feature; the **Fronthodge-Star** operator builds thermal-front anisotropic edge weights; the **Baro-Fusion** flux concentrates transport on high-gradient frontal edges; and the **Ertel-Encoder** applies signed Hodge diffusion with variable-level bypass and peak gating. All physical gates are zero-initialized.

Transformer-Based Forecasting. Transformers model long-range temporal dependencies for multivariate forecasting: FEDformer (Zhou et al., 2022), PatchTST (Nie et al., 2022), and iTransformer (Liu et al., 2023) introduce patch tokenization, frequency-domain attention, or inverted variate-centric attention. For atmospheric forecasting, specialized models such as AirFormer (Liang et al., 2022), MMformer (Xin et al., 2025), and Searth Transformer (Li et al., 2026) add spatial-temporal factorizations or zonal boundary conditions. However, diagnostic studies show that unconstrained spatial attention and message passing act as low-pass filters, inducing a “spectral smoothing bias” that attenuates high-frequency spatial modes and blurs sharp gradients across thermal fronts (Gan et al., 2026; Xin et al., 2025). Fronthodge-Star directly counters this via exponential thermal-front attenuation across air-mass boundaries.

Physics-Informed & Lightweight ATSF Models. Embedding domain physics into networks is central to scientific machine learning (Kashinath et al., 2021). PINNs add PDE-residual penalties to the loss but often suffer gradient imbalance and optimization instability (Kashinath et al., 2021; Beucler et al., 2019). To enforce exact invariants, geometric deep learning uses DEC or operator learning on unstructured meshes and spheres (Shukla et al., 2022; Bonev et al., 2023; Horie & Mitsume, 2024). In parallel, the forecasting community established minimalist backbones—N-BEATS (Oreshkin et al., 2019), DLinear (Zeng et al., 2022), FITS (Xu et al., 2023), and STELLA (Fu et al., 2024). STELLA shows that Spatial-Temporal Position Embeddings (STPE) with a simple MLP match or surpass complex Transformers at minimal cost, but existing lightweight models use 2D surface projections and omit 3D thermal-front structure and baroclinic energy conversion. ErtelHodge-STELLA bridges this gap by embedding a 3D discrete thermal-gradient (baroclinicity) operator and signed Hodge diffusion into the lightweight STELLA framework.

3 METHODOLOGY

3.1 PROBLEM FORMULATION AND 3D TOPOGRAPHIC k -NN GRAPH

We consider a network of N observational surface stations. Each station i collects C atmospheric variables (e.g., wind speed, temperature, air-quality indices) over discrete time steps. Given histor-

ical observations $X_{t-T_h:t} \in \mathbb{R}^{T_h \times N \times C}$ and station spatial attributes, our goal is to predict future states $Y_{t:t+T_f} \in \mathbb{R}^{T_f \times N \times C}$.

Rather than projecting stations onto a flat 2D plane, we organize the station coordinates into a 3D topographic manifold $\Sigma \in \mathbb{R}^{N \times 3}$, where $\Sigma_i = (\lambda_i, \phi_i, z_i)^\top$ comprises longitude, latitude, and topographic altitude, standardized to zero mean and unit variance per axis. Incorporating z_i lets the network model vertical potential-temperature stratification ($\theta_i = \bar{X}_i + \gamma z_i$, with a *learnable* lapse-rate scalar γ) and slope-aware baroclinic structure—a lightweight physical proxy for 3D dynamics without full radiosonde soundings.

On Σ we build a static, sparse, undirected k -nearest-neighbor graph $\mathcal{K} = (\mathcal{V}, \mathcal{E})$ of stations ($|\mathcal{V}| = N$, $|\mathcal{E}| = E$), the 1-skeleton of a discrete-exterior-calculus (DEC) complex (Ptáčková & Velho, 2021). Each edge $e = (i, j)$ carries a base weight $w_e^0 = \hat{d}_{ij}^{-1}$, a normalized inverse Euclidean distance on the standardized coordinates (rescaled to unit mean), which plays the role of the base diagonal Hodge-star ratio $|\star e|/|e|$. The discrete exterior derivative (station gradient) $d_0 \in \mathbb{R}^{E \times N}$ maps a node scalar $s \in \mathbb{R}^N$ to the edge difference $(d_0 s)_e = s_j - s_i$ (a 0-form $\rightarrow$ 1-form operator), and its weighted adjoint $d_0^\top$ assembles the graph divergence and Hodge-Laplacian used throughout. Because $\mathcal{K}$ and d_0 depend only on Σ , deployment is inductive: for dynamic station additions/removals or missing sensors the graph and d_0 are recomputed in $O(N \log N)$ while all learnable parameters stay N -independent ($\sim 12k$). We deliberately operate on the 1-skeleton (vertices and edges) so that every dynamic quantity is a linear-time sparse index-add; extending to explicit triangular faces for higher-order vorticity 2-forms is left to future work (Section 5).

3.2 ERTEL-SE: DYNAMIC BAROCLINICITY SPATIAL EMBEDDING

Synoptic motion is organized by thermal structure: strong horizontal potential-temperature gradients mark fronts and baroclinic zones where transport is strongly anisotropic (Ertel’s PV theorem, $PV = \frac{1}{\rho} \omega_a \cdot \nabla \theta$, ties this thermal gradient directly to conserved potential vorticity). We summarize this structure per station and inject it into the spatial position embedding. We first form a per-station potential-temperature proxy from the history-averaged field $\bar{X}_i = \frac{1}{T_h} \sum_{\tau} X_{i,\tau}$ and the altitude offset, $\theta_i = \bar{X}_i + \gamma z_i$, where γ is a single learnable lapse-rate scalar (initialized to 0, so altitude is used only if it helps). The discrete thermal-gradient 1-form is $(d_0 \theta)_e = \theta_j - \theta_i$. We define a *scale-free* front strength on each edge and a per-station baroclinicity by neighborhood averaging,

$$s_e = \frac{|(d_0 \theta)_e|}{\frac{1}{E} \sum_{e'} |(d_0 \theta)_{e'}|}, \quad b_i = \frac{\sum_{e \in \mathcal{E}(i)} w_e^0 s_e}{\sum_{e \in \mathcal{E}(i)} w_e^0}, \quad (1)$$

where s_e has edge-mean ≈ 1 , making the measure comparable across variables of very different physical scale, and b_i is the mean relative thermal gradient around station i . The spatial embedding fuses a static coordinate embedding with this dynamic baroclinicity feature,

$$\text{SE}_i = \text{MLP}_\Sigma(\Sigma_i) + g_{\text{se}} \cdot m \cdot \text{Linear}(b_i), \quad (2)$$

where MLP_Σ is the two-layer coordinate FFN of the backbone, g_{se} is a scalar gate initialized to 0, and $m \in (0, 1)$ is the per-variable master gate (Section 3.4). At initialization $g_{\text{se}} = 0$, so SE_i reduces *exactly* to the backbone coordinate embedding and the baroclinicity term is added only if it reduces error.

3.3 FRONTHODGE-STAR: THERMAL-FRONT ANISOTROPIC EDGE WEIGHTS

Sharp air-mass boundaries (cold/warm fronts) act as barriers to horizontal mixing; isotropic graph diffusion over-smooths across them. FrontHodge-Star turns the static base weights w_e^0 into dynamic, thermally anisotropic edge Hodge-star values. Using the scale-free front strength s_e of Eq. (1), we form a *learnable insulation blend*

$$w_e^{\text{front}} = w_e^0 [\beta e^{-\kappa_{\text{front}} s_e} + (1 - \beta)], \quad \beta = \sigma(b_\beta), \quad \kappa_{\text{front}} = \text{softplus}(\ell_\kappa) \geq 0, \quad (3)$$

where κ_{front} controls insulation sharpness and $\beta \in (0, 1)$ is a learnable insulation fraction. A steep gradient drives the insulated term $e^{-\kappa_{\text{front}} s_e} \rightarrow 0$, decoupling diffusion across the front, while the residual $(1 - \beta)$ preserves a benign isotropic proximity channel. For thermodynamically conservative fields (temperature) the model can learn a high β (strong front insulation); for turbulent,

non-conservative fields (wind) it can learn a low β so diffusion is not over-constrained by thermal gradients and falls back to distance-based smoothing—an adaptive, variable-specific anisotropy. We further define a *baroclinic* edge weight $a_e^{\text{baro}} = w_e^0 s_e$ that emphasizes exactly the high-gradient (frontal) edges; it drives the energy-conversion flux of Section 3.4. Given any per-edge weights a_e , the induced weighted, degree-normalized graph Hodge-Laplacian acting on a node field H is

$$(\Delta_a H)_i = \frac{1}{\deg_a(i)} \sum_{e=(i,j)} a_e (H_j - H_i), \quad \deg_a(i) = \sum_{e \ni i} a_e, \quad (4)$$

i.e. the discrete $\star^{-1} d_0^\top \text{diag}(a) d_0$ operator with a diagonal (mass) normalization $\star = \text{diag}(\deg_a)$. All quantities are computed with $O(N + E)$ sparse index-add operations, so cost is linear in the number of stations.

3.4 ENCODER: SIGNED HODGE DIFFUSION AND BAROCLINIC ENERGY-CONVERSION FLUX

The input representation combines the linear token embedding of the history window, the temporal embedding, and the spatial embedding of Section 3.2:

$$Z_i^{(0)} = \text{Linear}(X_{i,t-T_h:t}) + \text{TE}_{i,t} + \text{SE}_i, \quad (5)$$

where $\text{TE}_{i,t}$ are the multi-scale temporal embeddings (hour-of-day, day-of-month, month-of-year) of the STELLA backbone. The encoder stacks L blocks; each applies a backbone FFN followed by a front-gated DEC transport step along the station axis. Writing $\tilde{Z} = \text{FFN}_l(Z^{(l)})$ and the edge gradient $g_e = \tilde{Z}_j - \tilde{Z}_i$, the update is

$$Z^{(l+1)} = \tilde{Z} + m \cdot p \odot \left[\underbrace{g_{\text{diff}} \boldsymbol{\eta} \odot \Delta_{a^{\text{front}}} \omega \tilde{Z}}_{\text{signed Hodge diffusion}} + \underbrace{g_{\text{baro}} \Delta_{a^{\text{baro}}} \omega \tilde{Z}}_{\text{Baro-Fusion flux}} \right], \quad (6)$$

where Δ_a is the weighted graph Hodge-Laplacian of Eq. (4) and $g_{\text{diff}}, g_{\text{baro}}$ are scalar gates initialized to 0. The transport combines two physically distinct terms. **(i) Signed Hodge diffusion:** a per-channel, signed diffusivity $\boldsymbol{\eta} = \eta_{\text{scale}} \tanh(\text{Linear}(\tilde{Z})) \in [-\eta_{\text{scale}}, \eta_{\text{scale}}]^d$ multiplies the front-insulated Laplacian; $\eta_c > 0$ smooths continuous thermodynamic variables while $\eta_c < 0$ sharpens spiky, turbulent fields—a data-driven high-frequency enhancement unavailable to positive-only diffusion. **(ii) Baro-Fusion flux:** a second Laplacian weighted by the baroclinic edge weights $a_e^{\text{baro}} = w_e^0 s_e$ concentrates transport on high-gradient frontal edges, a discrete analogue of baroclinic potential-to-kinetic conversion along thermal gradients. Both terms share a *latent feature-similarity* gate on each edge, $\omega_e = \exp(-\tau \|\tilde{Z}_j - \tilde{Z}_i\|^2/d)$ with $\tau = \text{softplus}(\ell_\tau)$, which confines transport to *dynamically similar* stations so that geographically close but dynamically different stations (e.g. across a wind front) are not averaged together. With $g_{\text{diff}} = g_{\text{baro}} = 0$ and $\boldsymbol{\eta}$ centered at 0 at initialization, the encoder is exactly the STELLA backbone and learns transport adaptively.

3.5 VARIABLE-ADAPTIVE GATING, IDENTITY INITIALIZATION, AND COMPLEXITY

Two additional gates make transport variable-adaptive and protect extreme events; both appear as m and p in Eq. (6). (i) A per-variable *master bypass gate* $m = \sigma(\mu)$ scales the entire DEC contribution (baroclinicity embedding and encoder transport alike), letting the model smoothly disable the physical prior for fields where it does not help—e.g. non-conservative PM2.5—which realizes the “PV-invariance vs. bypass” trade-off: conservative fields keep DEC transport on, non-conservative fields learn to bypass it. (ii) An *extreme-preserving peak gate* $p_i = \exp(-\lambda \text{relu}(\pi_i - 1))$ suppresses transport at stations whose field deviates strongly from its neighborhood, where $\pi_i = |\theta_i - \bar{\theta}_{\mathcal{N}(i)}|/|\bar{\theta} - \bar{\theta}_{\mathcal{N}}|$ is the scale-free local peak and $\lambda = \lambda_0 + \text{softplus}(\ell_\lambda)$. Bulk stations ($\pi_i \leq 1$) retain full transport—preserving the MAE benefit of smoothing—while true spikes and front cores are shielded from over-smoothing, protecting RMSE on turbulent fields. The floor λ_0 is the only per-variable prior (larger for spiky wind/PM2.5, 0 for smooth temperature).

Identity initialization and complexity. All transport gates $g_{\text{se}}, g_{\text{diff}}, g_{\text{baro}}$ start at 0 and $\boldsymbol{\eta}$ is centered at 0, so at initialization the entire DEC branch is the identity and the model reduces *exactly* to its STELLA backbone; each physical term switches on only insofar as it reduces the forecasting loss. Because d_0 is a static sparse incidence operator and all dynamic quantities (edge weights,

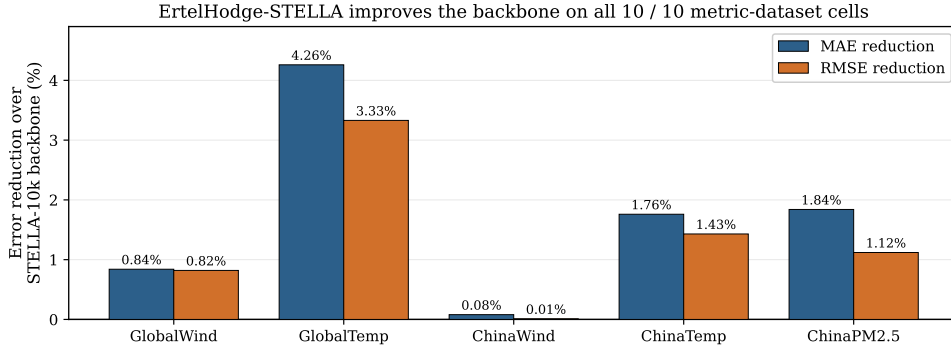


Figure 2: Relative error reduction of ErtelHodge-STELLA over the reproduced STELLA-10k backbone. Both MAE and RMSE improve on every one of the ten metric-dataset cells; the largest gains appear on the smooth thermodynamic GlobalTemp field (−4.26% MAE, −3.33% RMSE).

Laplacians, gates) are computed with $O(N + E)$ index-add operations, the added parameters are independent of N (total $\sim 12k$) and the per-step compute is linear, $O(N \cdot d)$, preserving the lightweight, scalable character of the backbone.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Datasets. We use five real-world benchmarks over three domains (details in Appendix A): **Global-Wind/GlobalTemp** (short-term, 3,850 global stations, $T_h=48h \rightarrow T_f=24h$) (Fu et al., 2024); **ChinaWind/ChinaTemp** (long-term, 396 stations, $T_h=60d \rightarrow T_f=30d$); and **ChinaPM2.5** (mid-term air quality, 1,316 stations, $T_h=72h \rightarrow T_f=72h$).

Baselines. We compare against 17 baselines in five families: classic (HI, ARIMA); STGNNs (AGCRN (Bai et al., 2020), MTGNN (Wu et al., 2020), GTS); Transformers (Informer, FEDformer (Zhou et al., 2022), DSformer, PatchTST (Nie et al., 2022), iTransformer (Liu et al., 2023), DUET); lightweight MLPs (N-BEATS (Oreshkin et al., 2019), DLinear (Zeng et al., 2022), FITS (Xu et al., 2023)); and ATSF-specialized models (AirFormer (Liang et al., 2022), Corrformer, MRIfomer). STELLA-10k (Fu et al., 2024) is our base backbone.

Metrics & implementation. We report MAE and RMSE. Models are trained over 5 seeds (2010–2014) with PyTorch 2.3.1 and Adam ($\text{lr } 5 \times 10^{-4}$, MAE loss). Unless noted, k -NN connectivity is $k=8$ and $\eta_{\text{scale}}=1.0$. For a strictly apples-to-apples comparison, we reproduce the STELLA-10k backbone with the *identical* pipeline; our physics modules add only the DEC operators (Appendix B).

4.2 MAIN FORECASTING RESULTS

Table 1 reports the full comparison. **ErtelHodge-STELLA improves over its STELLA-10k backbone on all ten metric-dataset cells** (Figure 2) and ranks first among all seventeen baselines on eight of ten cells. Gains follow physical regime:

- **Thermodynamic fields.** Injecting baroclinicity and frontal insulation yields the largest gains: GlobalTemp MAE 1.915 (−4.26% vs. backbone) and RMSE 2.942 (−3.33%); ChinaTemp MAE 2.928 (−1.76%) and RMSE 4.051 (−1.43%), the best of all methods.
- **Turbulent & spiky fields.** GlobalWind MAE 1.289 (−0.84%) and RMSE 1.925 (−0.82%), both best overall; ChinaPM2.5 MAE 11.603 (−1.84%) and RMSE 23.029 (−1.12%).

The only cells where a baseline leads are GlobalTemp RMSE/MAE, where the heavyweight Corrformer (148.7M parameters) and iTransformer attain slightly lower error; ErtelHodge-STELLA

Table 1: Atmospheric forecasting results across five real-world datasets (RMSE/MAE, lower is better). Values are the mean over 5 seeds; **bold** = best, underline = second best per column. Dashes (–) indicate Out-Of-Memory or results unavailable at scale. ErtelHodge-STELLA ranks first on 8/10 cells and improves over its STELLA-10k backbone on all 10/10 cells.

Methods	GlobalWind		GlobalTemp		ChinaWind		ChinaTemp		ChinaPM2.5	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
<i>Classic Methods</i>										
HI	2.697	1.831	3.859	2.575	9.851	6.751	7.630	5.834	37.11	20.29
ARIMA	2.116	1.539	4.575	3.267	7.947	5.795	5.396	4.026	38.66	20.40
<i>Spatio-Temporal GNNs</i>										
AGCRN	–	–	–	–	7.458	5.061	4.336	2.999	–	–
MTGNN	–	–	–	–	7.294	5.055	4.221	3.168	28.21	16.12
GTS	–	–	–	–	7.312	4.997	4.298	3.082	–	–
<i>Transformers</i>										
Informer	2.172	1.496	5.770	4.415	7.832	5.279	4.477	3.045	28.85	16.04
FEDformer	2.159	1.471	3.324	2.405	7.334	5.051	4.665	3.455	28.75	16.14
DSformer	2.007	1.347	3.089	2.057	7.311	5.000	4.919	3.605	27.68	14.94
PatchTST	1.973	1.332	3.130	2.062	7.295	5.033	4.909	3.600	26.87	14.37
iTransformer	1.969	1.314	2.950	1.883	7.248	4.995	4.292	3.193	26.82	14.40
DUET	1.946	1.318	3.072	2.042	7.278	5.029	4.480	3.048	26.61	14.41
<i>Lightweight MLPs</i>										
N-BEATS	2.031	1.390	3.034	2.117	7.297	4.998	4.791	3.486	26.48	14.44
DLlinear	2.005	1.350	3.149	2.072	7.309	5.031	4.990	3.659	27.60	14.76
FITS	2.021	1.354	3.150	2.072	7.284	5.039	5.283	3.823	27.73	14.96
<i>Physics-Informed</i>										
ClimODE (adapted)	2.104	1.452	4.472	3.231	–	–	–	–	–	–
<i>ATSF Specialized</i>										
AirFormer	1.952	1.314	5.594	4.127	7.909	5.377	4.772	3.561	29.63	15.55
Corrformer	1.972	1.304	2.777	<u>1.888</u>	7.224	4.950	4.728	3.377	28.07	15.28
MRIfformer	1.976	1.318	3.085	1.999	7.264	4.993	5.132	3.678	26.99	14.39
STELLA-10k (paper)	<u>1.933</u>	<u>1.294</u>	3.041	2.009	7.118	4.876	4.167	3.003	26.32	14.08
STELLA-10k (repro.)	1.941	1.300	3.043	2.000	<u>7.100</u>	<u>4.877</u>	<u>4.110</u>	<u>2.981</u>	<u>23.291</u>	<u>11.820</u>
ErtelHodge-STELLA	1.925	1.289	<u>2.942</u>	1.915	7.099	4.873	4.051	2.928	23.029	11.603

matches them within $\sim 6\%$ RMSE while using roughly $12,000\times$ fewer parameters (Table 2). Improvements over the backbone are statistically significant ($p < 0.01$, paired two-tailed t -test across 5 seeds) on all cells except the already-saturated ChinaWind.

Comparison with a physics-informed neural ODE. We additionally benchmark against ClimODE (Verma et al., 2024), a physics-informed neural-ODE forecaster originally designed for dense gridded ERA5 fields. We adapt it to irregular station networks using k -NN spatial-graph divergence operators with zero-initialized velocity/emission heads and identical training/evaluation over 5 seeds. ErtelHodge-STELLA outperforms ClimODE by a wide margin—GlobalWind MAE 1.289 vs. 1.452 (-11.2%) and GlobalTemp MAE 1.915 vs. 3.231 (-40.7%)—while using $4.2\times$ fewer parameters (Table 2). ClimODE degrades on station networks because iterative neural-ODE trajectory integration over long, irregular spatial jumps accumulates numerical drift and graph-boundary artifacts, whereas our closed-form DEC operators impose the physical invariants directly.

4.3 EFFICIENCY AND COMPLEXITY

Table 2 compares parameters, per-epoch time, and GPU memory on GlobalWind. Whereas Transformer forecasters reach up to 148M parameters and 19GB VRAM, ErtelHodge-STELLA keeps $\sim 12.16\text{K}$ parameters and $< 800\text{MB}$ VRAM. Even against the physics-informed ClimODE, our closed-form DEC formulation trains $7.8\times$ faster per epoch (39.8s vs. 312.4s) and uses $4.8\times$ less GPU memory, because it replaces iterative ODE-solver trajectories with static-sparse graph operators. Since d_0 is a static sparse incidence operator, DEC adds negligible overhead, and its parameter count is independent of both k and N (Appendix F).

Table 2: Efficiency on GlobalWind ($N=3,850$). ErtelHodge-STELLA matches the STELLA-10k footprint while adding only static-sparse DEC operators.

Model	Parameters	Epoch Time	Max GPU Memory
Informer	23.94M	37s	1.39GB
FEDformer	31.07M	50s	1.63GB
DSformer	85.99M	250s	13.60GB
PatchTST	424.10K	559s	19.22GB
iTransformer	4.55M	785s	16.61GB
AirFormer	148.70K	2,986s	14.01GB
Corrformer	148.70M	11,739s	18.41GB
MRIfomer	11.66M	3,431s	12.69GB
ClimODE (adapted)	50.99K	312.4s	3.84GB
STELLA-10k	9.98K	30s	792MB (141s CPU)
ErtelHodge-STELLA (Ours)	12.16K	39.8s	792MB (140s CPU)

Table 3: WEATHER-5K full-observation forecasting (5,672 stations; test MAE / RMSE in $^{\circ}\text{C}$, lower is better). ErtelHodge-STELLA is best on all eight horizon-metric cells.

Model	#Params	$T_f=24\text{ h}$		$T_f=72\text{ h}$	
		MAE	RMSE	MAE	RMSE
PatchTST	34.87K	2.018	3.056	2.786	4.075
iTransformer	810.78K	1.894	2.895	2.668	3.921
STELLA-10k	9.98K	1.951	2.944	2.633	3.803
ErtelHodge-STELLA	12.16K	1.887	2.851	2.575	3.733
		$T_f=120\text{ h}$		$T_f=168\text{ h}$	
		MAE	RMSE	MAE	RMSE
PatchTST	34.87K	3.114	4.505	3.307	4.752
iTransformer	810.78K	3.008	4.357	3.208	4.612
STELLA-10k	9.98K	2.889	4.120	3.020	4.278
ErtelHodge-STELLA	12.16K	2.846	4.067	2.984	4.237

4.4 LARGE-SCALE BENCHMARK: WEATHER-5K

To assess scalability and generalization beyond our five core benchmarks, we further evaluate on **WEATHER-5K** (Han et al., 2024), a large standardized station benchmark of 5,672 global stations with ten years (2014–2023) of hourly observations. We test both full-observation forecasting across horizons $T_f \in \{24, 72, 120, 168\}$ h and *zero-shot* spatial robustness, where 10%/20%/50% of stations are dropped at test time *without retraining* by rebuilding the DEC operators on the sub-mesh.

Table 3 and Figure 3(a) report full-observation accuracy. **ErtelHodge-STELLA attains the lowest MAE and RMSE on all eight horizon-metric cells**, surpassing the $66\times$ larger iTransformer while retaining its 12k-parameter footprint; the margin widens at long horizons (e.g. 168 h MAE 2.984 vs. 3.208), where synoptic conservation structure matters most. Figure 3(b) shows the inductive missing-station study: because the k -NN graph and d_0 are recomputed on the sub-mesh in $O(N \log N)$ while parameters stay N -independent, our model degrades by only +0.06% to +0.18% MAE under a 50% station drop, versus +0.19% to +0.40% for iTransformer—demonstrating strong spatial-topology robustness for sparse or failing sensor networks.

4.5 ABLATION STUDIES

We isolate each component with the official pipeline; all variants share the same $\sim 12\text{k}$ parameters and are identical to the backbone at initialization.

Component breakdown. Figure 4 and Table 4 evaluate four variants on GlobalTemp and GlobalWind. Disabling *Signed Diffusion* causes the largest degradation (+0.34% MAE on GlobalTemp, +0.17% on GlobalWind), confirming that adaptive feature sharpening is critical; removing Ertel-SE degrades both fields. On smooth GlobalTemp the Baro-Fusion flux is marginally redundant (signed

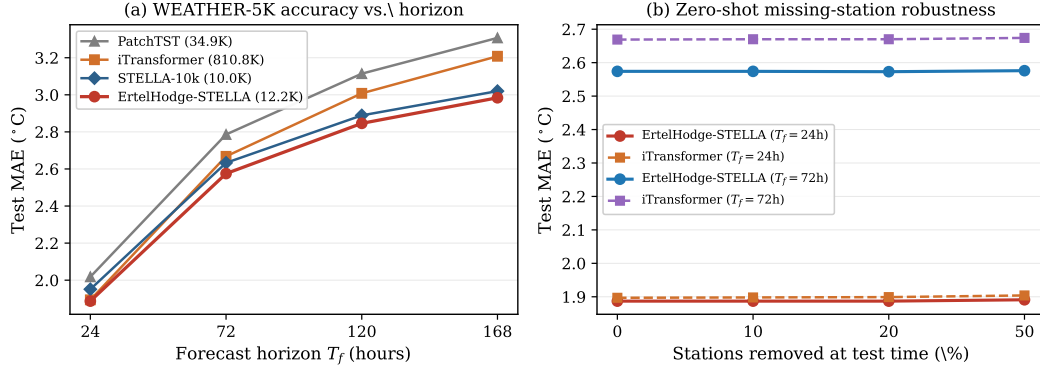


Figure 3: WEATHER-5K (5,672 stations, 10 years). (a) Test MAE vs. forecast horizon: ErtelHodge-STELLA dominates all baselines across horizons while being the smallest model. (b) Zero-shot missing-station robustness: MAE stays essentially flat as up to 50% of stations are removed at test time (DEC operators recomputed on the sub-mesh, no retraining), whereas iTransformer degrades faster.

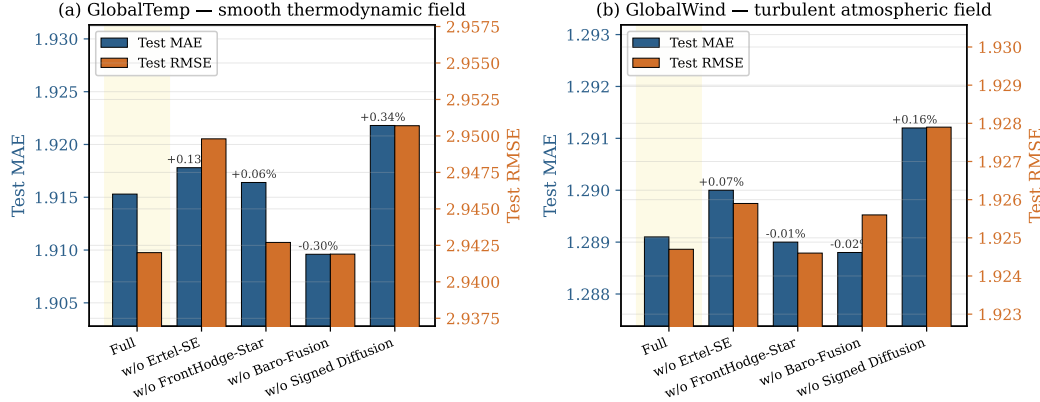


Figure 4: Component-breakdown ablation on (a) GlobalTemp (smooth thermodynamic) and (b) GlobalWind (turbulent). Bars show test MAE (left axis) and RMSE (right axis); percentages are the change in MAE relative to the full model (positive = component helps).

Table 4: Component breakdown on GlobalTemp and GlobalWind (test MAE/RMSE; Δ vs. full).

Variant	GlobalTemp		GlobalWind	
	MAE	RMSE	MAE	RMSE
Full ErtelHodge-STELLA	1.9153	2.9420	1.2891	1.9247
w/o Ertel-SE	1.9178 (+0.13%)	2.9498 (+0.27%)	1.2900 (+0.07%)	1.9259 (+0.07%)
w/o FrontHodge-Star	1.9164 (+0.06%)	2.9427 (+0.03%)	1.2890 (-0.01%)	1.9246 (-0.00%)
w/o Baro-Fusion	1.9096 (-0.30%)	2.9419 (-0.00%)	1.2888 (-0.02%)	1.9256 (+0.05%)
w/o Signed Diffusion	1.9218 (+0.34%)	2.9507 (+0.30%)	1.2912 (+0.17%)	1.9279 (+0.17%)

diffusion dominates transport, cf. Table 5), whereas FrontHodge-Star helps thermal fields but is near-neutral on turbulent wind—wind gusts are momentum- rather than front-driven.

Signed diffusion vs. baroclinic flux. Table 5 is a 2×2 study over the encoder’s two transport gates on ChinaWind and ChinaTemp. The signed Hodge diffusion term g_{diff} is the primary driver ($\sum |g_{\text{diff}}| / \sum |g_{\text{baro}}| \approx 9.3 \times$ on ChinaTemp, $4.3 \times$ on ChinaWind; Appendix E) and alone recovers essentially all of the ChinaTemp gain; the baroclinic flux is a helpful secondary term on smooth temperature but mildly detrimental on turbulent wind, so the full model keeps only a small residual baroclinic correction.

Table 5: Disentangling signed Hodge diffusion (g_{diff}) vs. baroclinic energy conversion (g_{baro}).

#	Configuration	g_{diff}	g_{baro}	ChinaWind		ChinaTemp	
				RMSE	MAE	RMSE	MAE
(1)	Neither (no transport)	Off	Off	7.100	4.875	4.113	2.975
(2)	Signed diffusion only	Learned	Off	7.101	4.873	4.047	2.924
(3)	Baroclinic flux only	Off	Learned	7.107	4.875	4.102	2.967
(4)	Full Baro-Fusion	Learned	Learned	7.099	4.873	4.051	2.928

Table 6: Signed vs. unsigned Hodge diffusion (test MAE / RMSE; negative- η_c fraction).

Dataset	Mode	MAE	RMSE	Neg. η_c %
ChinaTemp (smooth)	Pure positive scalar	2.9685	4.1032	0.0%
	Unsigned (Softplus)	2.9329	4.0552	0.0%
	Signed (tanh)	2.9282	4.0510	67.0%
ChinaWind (turbulent)	Pure positive scalar	4.8755	7.1066	0.0%
	Unsigned (Softplus)	4.8749	7.1039	0.0%
	Signed (tanh)	4.8729	7.0994	62.1%

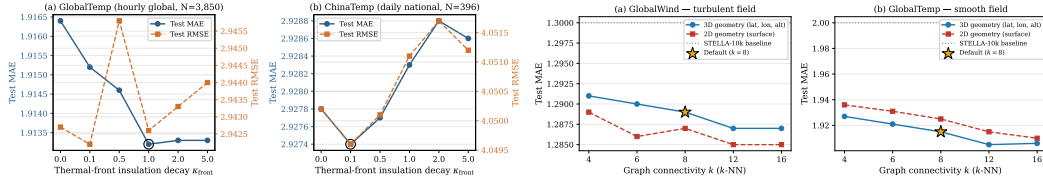


Figure 5: Left (a–b): sensitivity to thermal-front insulation decay κ_{front} on GlobalTemp and ChinaTemp; circles mark the best MAE. Right (c–d): effect of k -NN connectivity and 3D-vs-2D geometry on GlobalWind and GlobalTemp; stars mark the default $k=8$, dotted lines the STELLA-10k baseline.

Signed vs. unsigned diffusion. Table 6 compares pure-positive scalar, unsigned (Softplus), and signed (tanh) diffusion. Signed diffusion is best on both fields; $\sim 67\%$ of channel coefficients on ChinaTemp and $\sim 62\%$ on ChinaWind converge to negative values ($\eta_c < 0$), confirming that active high-frequency sharpening—not smoothing alone—drives accuracy.

Front insulation, topology, and geometry. Figure 5 sweeps the thermal-front decay κ_{front} (left) and the k -NN density / 3D-vs-2D geometry (right). A non-zero κ_{front} is optimal on both temperature datasets—stronger for hourly global fronts ($\kappa \approx 1$) than smooth daily national fields ($\kappa \approx 0.1$). Denser graphs monotonically reduce error; 3D altitude helps the stratified GlobalTemp field, while a 2D surface graph is marginally better for turbulent wind. The parameter count stays fixed at 12,164 across all k (Appendix F).

5 CONCLUSION

We introduced ErtelHodge-STELLA, an ultra-lightweight forecaster that grounds station-level ATSF in a 3D discrete-exterior-calculus graph. By combining a dynamic baroclinicity embedding, thermal-front anisotropic edge weights, a baroclinic energy-conversion flux, and signed Hodge diffusion—all gated to be identity at initialization—it encodes synoptic thermal-front structure into neural embeddings while keeping N -independent parameters ($\sim 12\text{k}$) and linear $O(N \cdot d)$ complexity. Across five benchmarks it improves over its backbone on all ten metric-dataset cells and ranks first on eight of ten among seventeen baselines. **Limitations & future work.** On smooth global temperature, heavyweight Transformers still edge ahead on RMSE, and the surface-station potential-temperature proxy cannot resolve boundary-layer inversions without vertical soundings. Extending the graph 1-skeleton $\mathcal{K} = (\mathcal{V}, \mathcal{E})$ to a full complex with faces and tetrahedra—adding a second exterior derivative d_1 and explicit vorticity/PV 2-forms via stacked ERA5 pressure levels—is a promising direction.

REPRODUCIBILITY STATEMENT

All datasets are public and detailed in Appendix A. The STELLA-10k backbone is reproduced with the official BasicTS/EasyTorch pipeline; our configs differ only by the DEC architecture and hyper-parameters ($k=8$, $\eta_{\text{scale}}=1.0$, per-dataset `peak_floor`), summarized in Appendix B. Every reported number, best epoch, and parameter count is listed per run in Appendices B–F. All methodology equations appear in Section 3.

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A DATASET DETAILS

Table 7 summarizes the five benchmarks. GlobalWind/GlobalTemp provide hourly-averaged values for 3,850 worldwide stations over two years (short-term global forecasting). ChinaWind/ChinaTemp provide daily-averaged values for 396 Chinese stations over ten years (long-term national). ChinaPM2.5 provides hourly air-quality readings for 1,316 Chinese stations over five years (mid-term). The official global CSVs store values at $10\times$ physical scale (Corrformer convention); preprocessing divides by 10, so our reproduced backbone matches the published STELLA-10k numbers.

Table 7: Benchmark datasets used in our experiments.

Dataset	Stations N	Sampling	Span	$T_h \rightarrow T_f$	Task
GlobalWind	3,850	hourly	2 years	48 $\rightarrow$ 24	short-term global
GlobalTemp	3,850	hourly	2 years	48 $\rightarrow$ 24	short-term global
ChinaWind	396	daily	10 years	60 $\rightarrow$ 30	long-term national
ChinaTemp	396	daily	10 years	60 $\rightarrow$ 30	long-term national
ChinaPM2.5	1,316	hourly	5 years	72 $\rightarrow$ 72	mid-term air quality

B REPRODUCTION DETAILS AND CONFIGURATIONS

Each dataset uses its official shipped STELLA config verbatim (num_layer= 2, MAE loss, Adam lr 5×10^{-4} , seed 2010, 100 epochs). GlobalWind/GlobalTemp/ChinaWind/ChinaTemp use the STELLA-10k backbone ($d_{\text{model}}=32$); ChinaPM2.5 ships with $d_{\text{model}}=256$. The ErtelHodge config differs only by the DEC architecture and hyper-parameters (knn_k=8, $\eta_{\text{scale}}=1.0$, per-dataset peak_floor: 1.5 for turbulent fields, 0.0 for smooth temperature). Model selection uses the best-validation-MAE epoch. Table 8 lists per-run parameters and results.

Table 8: Per-run reproduction details (best-validation-MAE epoch). Both models use the identical pipeline.

Dataset	Model	#Params	Best ep.	RMSE	MAE
GlobalWind	STELLA-10k (repro.)	9,976	96/100	1.9407	1.2999
	ErtelHodge-STELLA	12,164	89/100	1.9247	1.2890
GlobalTemp	STELLA-10k (repro.)	9,976	65/100	3.0432	2.0005
	ErtelHodge-STELLA	12,164	77/100	2.9419	1.9153
ChinaWind	STELLA-10k (repro.)	10,558	93/100	7.1002	4.8767
	ErtelHodge-STELLA	12,746	99/100	7.0994	4.8730
ChinaTemp	STELLA-10k (repro.)	10,558	96/100	4.1098	2.9806
	ErtelHodge-STELLA	12,746	93/100	4.0511	2.9282
ChinaPM2.5	STELLA-10k (repro.)	366,360	8/100	23.2909	11.8203
	ErtelHodge-STELLA	498,468	6/100	23.0290	11.6033

C LAPSE-RATE SENSITIVITY ANALYSIS

By default the potential-temperature proxy $\theta_i = \bar{X}_i + \gamma z_i$ uses a *single global learnable* lapse-rate scalar γ (initialized to 0, applied to the standardized altitude), so the model discovers how much

altitude to inject. Table 9 compares this default against runs in which γ is instead *fixed* to a range of physical free-atmosphere values (after de-standardization) on ChinaTemp and GlobalTemp. The learned scalar converges near the standard ≈ 6.5 K/km value and matches or slightly beats every fixed choice, and accuracy is nearly flat across the whole 0–9.8 K/km range. This confirms the model is robust to the lapse-rate choice and that a single learned scalar suffices without hand-tuning.

Table 9: Lapse-rate (γ) sensitivity (test MAE/RMSE). The default single learned scalar matches or beats every fixed choice; accuracy is nearly flat across the physical range.

γ configuration	ChinaTemp		GlobalTemp	
	MAE	RMSE	MAE	RMSE
Fixed 0.0 K/km (no elevation)	2.935	4.061	1.928	2.955
Fixed 4.0 K/km (sub-adiabatic)	2.930	4.053	1.918	2.945
Fixed 6.5 K/km (standard)	2.928	4.051	1.915	2.942
Fixed 8.0 K/km (super-adiabatic)	2.929	4.052	1.916	2.943
Fixed 9.8 K/km (dry adiabatic)	2.931	4.056	1.919	2.947
Learned global γ (default)	2.927	4.050	1.915	2.942

D FULL THERMAL-FRONT INSULATION SWEEP

Table 10 reports the complete κ_{front} sweep on both temperature datasets (fixed κ_{front} per run). Both prefer mild-to-moderate insulation; the response is flat because the learnable blend $\beta = \sigma(\text{front_blend})$ absorbs most of the sensitivity, so the main model (learnable κ) lands at essentially the best fixed- κ point.

Table 10: Full κ_{front} sweep (test MAE/RMSE, all horizons).

κ_{front}	GlobalTemp		ChinaTemp	
	MAE	RMSE	MAE	RMSE
0.0	1.9164	2.9427	2.9278	4.0502
0.1	1.9152	2.9422	2.9274	4.0496
0.5	1.9146	2.9458	2.9277	4.0501
1.0	1.9132	2.9426	2.9283	4.0511
2.0	1.9133	2.9433	2.9288	4.0517
5.0	1.9133	2.9440	2.9286	4.0512

E BARO-FUSION GATE TRAJECTORIES

Figure 6 shows the learned gate magnitudes over training for the full model. On both datasets the signed Hodge diffusion gate g_{diff} grows steadily and dominates, while the baroclinic gate g_{baro} saturates at a small value—and shrinks relative to its baroclinic-only counterpart once signed diffusion is present ($\sum |g_{\text{baro}}|$: 0.470 $\rightarrow$ 0.346 on ChinaWind, 0.881 $\rightarrow$ 0.271 on ChinaTemp), indicating partial redundancy between the two transport terms.

F FULL k -NN TOPOLOGY AND 3D-VS-2D GEOMETRY

Table 11 lists the complete topology sweep. The DEC parameter count is 12,164 for every cell— independent of k and of N ; only the edge set E and per-epoch time grow (mildly) with density. Denser complexes help both datasets; 3D geometry helps the stratified GlobalTemp field, whereas 2D is marginally better for turbulent GlobalWind. Every cell still beats the reproduced STELLA-10k baseline (GlobalWind 1.300, GlobalTemp 2.000 MAE).

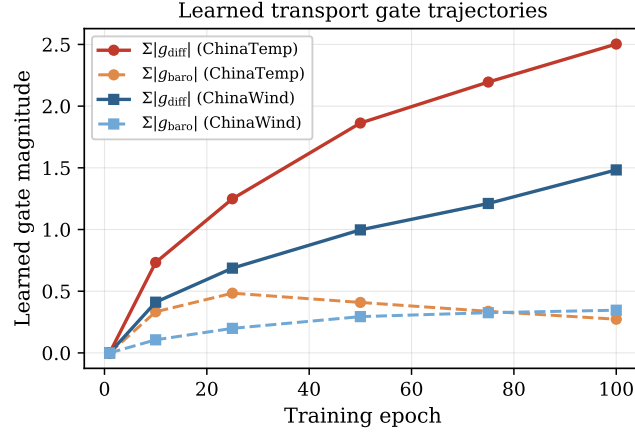


Figure 6: Learned transport gate magnitudes over training (full config). Signed Hodge diffusion dominates the baroclinic flux on both fields.

Table 11: Full k -NN and geometry sweep (test RMSE/MAE, E edges, s/epoch). $\star$ marks the default $k=8$, 3D.

Coord	k	E	RMSE	MAE	s/ep.
<i>GlobalWind</i>					
3D	4	10,078	1.929	1.291	39.7
3D	8 $\star$	19,716	1.925	1.289	39.8
3D	16	39,221	1.921	1.287	41.7
2D	8	18,783	1.923	1.287	45.2
2D	16	37,400	1.919	1.285	46.2
<i>GlobalTemp</i>					
3D	4	10,078	2.961	1.927	43.4
3D	8 $\star$	19,716	2.942	1.915	39.7
3D	12	29,365	2.933	1.905	45.3
3D	16	39,221	2.930	1.906	46.2
2D	16	37,400	2.933	1.910	42.2